Title: Spatially-aware CatBoost Benchmark for Predicting *Listeria* Presence in U.S. Soil Samples
Team Name: Decaying Beta-Amyloid
Repository Link: https://github.com/tkinley/Listeria-CatBoost-Predictor

Problem and operational value: Soil can act as a long-term reservoir for *Listeria* spp., creating downstream risk for produce and other foods. We benchmark a predictive model that estimates the probability of *Listeria* presence in U.S. soil samples using soil chemistry, climate, land-use composition, and geographic coordinates. The goal is to support targeted surveillance and sampling, prioritizing locations with elevated predicted risk under limited monitoring resources.
Data: We used the Cornell Food Safety ML Repository "*Listeria* in soil" dataset (cleaned file ListeriaSoil_clean.csv with corresponding metadata). The raw outcome is the number of *Listeria* isolates obtained per sample. We constructed a binary label: presence = 1 if isolates > 0, absence = 0 otherwise. The dataset includes environmental predictors spanning soil properties (nutrients and metals), meteorological variables, land cover percentages, and latitude/longitude.

Methods: *Locked evaluation protocol*; To avoid overly optimistic evaluation caused by spatial autocorrelation (nearby samples sharing similar environmental conditions), we used a locked, spatially aware cross-validation protocol. Samples were assigned to spatial groups using fixed latitude/longitude grid cells (0.25°), and splits were performed by group using StratifiedGroupKFold (5 folds, seed 42). Model selection and comparisons were performed under this locked protocol.

Model : We trained a CatBoost gradient-boosted tree classifier on the full set of predictors. CatBoost was selected for strong performance on tabular data and robust handling of nonlinear feature interactions.
Metrics and threshold policy: We reported classification metrics aligned with the competition rubric outlined in Table 1. We selected an operating threshold by maximizing F1 on out-of-fold (OOF) predictions under the locked evaluation.

Results : Under the locked spatial evaluation, the model achieved strong discrimination and balanced classification performance:
Table 1: Metrics for benchmarking (Locked threshold (F1-optimized) :0.475)

	ROC AUC	PR AUC	F1	Sensitivity	Specificity
Mean value	0.936	0.932	0.872	0.897	0.839
Std dev	0.022088	0.027356	0.034788	0.035616	0.040194

These results indicate high true-positive capture while maintaining reasonable control of false positives.

Interpretability: Feature-importance analysis highlighted a small set of dominant predictors of *Listeria* occurrence in soil, including soil chemistry (notably sodium and trace metals such as molybdenum), moisture, and geographic coordinates. These variables plausibly capture combinations of edaphic properties and broader geochemical/ecological gradients that influence *Listeria* survival and biotic interactions in soil, consistent with evidence that soil physiochemistry is a major determinant of *L. monocytogenes* survival. Given *Listeria*'s ability to survive and adapt under salt stress, sodium may act as a proxy for salinity-linked habitats permissive for persistence,

while molybdenum may represent the soil geochemistry and redox/nutrient status rather than a direct mechanistic driver. Accordingly, we interpret feature importance as predictive association, not causation.

Calibration and actionability:
Probability calibration: Beyond discrimination, operational monitoring benefits from well-calibrated risk scores. We calibrated OOF predicted probabilities using isotonic regression, improving probability accuracy: Brier score (uncalibrated): 0.1008; Brier score (isotonic calibrated): 0.0945. A reliability diagram (Figure 2) shows an improved agreement between predicted risk and observed positivity after calibration.

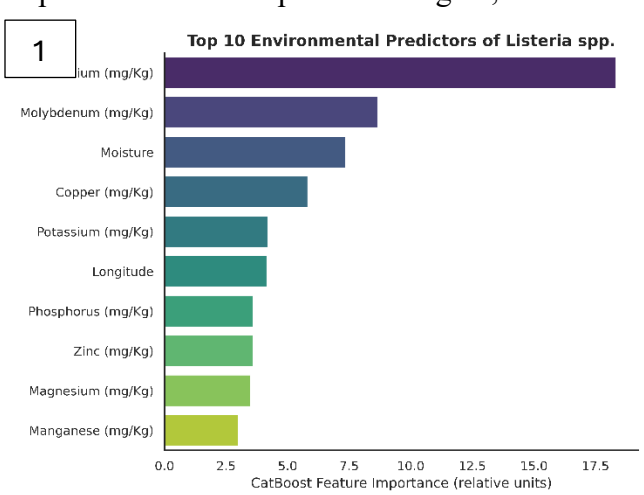
Decision policies for real monitoring
We illustrate an actionable capacity-based sampling policy using the calibrated risk score. When selecting only the highest-risk locations:

- Top 10% predicted risk: 98.4% observed positivity (vs 50.0% overall)
- Top 20% predicted risk: 96.8% observed positivity
- Top 30% predicted risk: 95.2% observed positivity

This enrichment demonstrates that the model can substantially increase sampling efficiency when surveillance resources are constrained.

Reproducibility and system requirements:
All code is organized to be reproducible via a locked evaluation configuration, saved OOF predictions, and exported figures and metrics (feature importance, ROC/PR curves, confusion matrix, and calibration curve). The submission includes system requirements (CPU/RAM) and a single-command run specification in the repository README.

Limitations
Performance is estimated using spatial group cross-validation, which reduces leakage but can still be sensitive to group definition and sample density. The dataset includes only available measured covariates; unmeasured factors (management practices) could influence presence. Feature importance indicates predictive signal, not causal mechanisms.



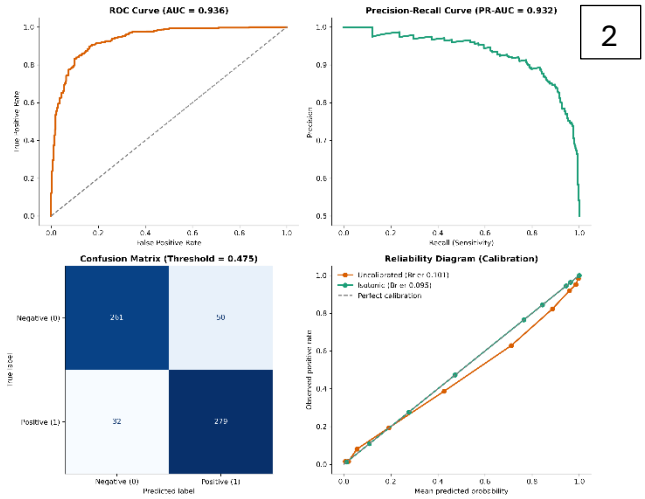


Figure 1: Top 10 feature importances (CatBoost); *Figure 2*: Combined panel: ROC curve, PR curve, confusion matrix at locked threshold, and reliability diagram (uncalibrated vs isotonic calibrated) .

References:

Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A. V., and Gulin, A. (2018). CatBoost: Unbiased boosting with categorical features. *NeurIPS*.

Pedregosa, F., Varoquaux, G., Gramfort, A., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.